

CAREER INTELLIGENCE AS A RETENTION STRATEGY: UNCOVERING PROMOTION GAPS, CAREER STAGNATION DYNAMICS, AND PROACTIVE TALENT INTERVENTIONS

An Unsupervised Machine Learning Framework & Decision Intelligence System for Enterprise Workforce Management

UNIFIED MENTOR MACHINE LEARNING INTERNSHIP PROJECT REPORT

Author / Intern:	Sarupya Guha
Intern ID:	UMID150826105831
Role:	Machine Learning Engineering Intern
Organization:	Unified Mentor
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DECLARATION OF AUTHENTICITY

I hereby declare that this project report titled "**Career Intelligence as a Retention Strategy: Uncovering Promotion Gaps, Career Stagnation Dynamics, and Proactive Talent Interventions for Palo Alto Networks**" submitted to **Unified Mentor** represents my original work carried out as part of the Machine Learning Internship Program.

All algorithms, data engineering pipelines, mathematical modeling, simulation engines, and Streamlit application interfaces documented in this report were developed in accordance with the project guidelines. Appropriate citations and disclaimers have been provided for all external methodologies and reference materials.

Sarupya Guha

Machine Learning Engineering Intern

Unified Mentor

Intern ID: UMID150826105831

TABLE OF CONTENTS

- **Chapter 1: Introduction and Problem Formulation**
 - 1.1 Organizational Background: Palo Alto Networks Workforce Context
 - 1.2 The Paradigm Shift: Reactive Turnover Prediction vs. Proactive Career Intelligence
 - 1.3 Core Research Objectives & Hypotheses
- **Chapter 2: Data Architecture and Domain Feature Engineering**
 - 2.1 Dataset Overview & Schema Profiling
 - 2.2 Mathematical Formulations for Derived Career Indicators
 - 2.2.1 Promotion Gap Ratio (\$PGR\$)
 - 2.2.2 Role Stagnation Index (\$RSI\$)
 - 2.2.3 Training Intensity Score (\$TIS\$)
 - 2.2.4 Manager Stability Indicator (\$MSI\$)
 - 2.2.5 Career Velocity (\$CV\$)
 - 2.3 Composite Promotion Gap Risk Scoring (\$PGRS\$)
 - 2.4 Retention Opportunity Index Formulation (\$ROI\$)
- **Chapter 3: Unsupervised Career Path Clustering & Archetype Discovery**
 - 3.1 Preprocessing Pipeline & Multi-Dimensional Standardization
 - 3.2 K-Means Clustering Formulation & Optimal \$K\$ Selection
 - 3.3 Hierarchical Agglomerative Validation
 - 3.4 Principal Component Analysis (PCA) Dimensionality Reduction
 - 3.5 Centroid Interpretation & 5 Career Archetypes
- **Chapter 4: Promotion Gap Risk & Retention Opportunity Modeling**
 - 4.1 Enterprise Stagnation Distribution
 - 4.2 Cross-Departmental Vulnerability Analysis
 - 4.3 High-Performer Retention Opportunity Priority Queue
- **Chapter 5: Managerial Continuity & Leadership Impact Dynamics**
 - 5.1 Managerial Tenure vs. Promotion Latency Correlation
 - 5.2 Stagnant Leadership Dyads & Structural Bottlenecks
 - 5.3 Team-Level Leadership Progression Signals
- **Chapter 6: Individual Career Simulator & Counterfactual What-If Lab**
 - 6.1 Simulation Engine Mathematical Mechanics
 - 6.2 Trajectory Shift Projection on Latent PCA Space

- 6.3 Prescriptive Next-Best-Action Decision Framework
 - **Chapter 7: Streamlit Enterprise Decision Intelligence Platform**
 - 7.1 System Architecture & UI/UX Design Principles
 - 7.2 Module Breakdown (Overview, Clustering, Gaps, Retention, Simulation, Explorer)
 - **Chapter 8: Conclusion, Financial ROI Model, and Policy Recommendations**
 - 8.1 Summary of Quantitative Impacts
 - 8.2 Enterprise Cost Avoidance ROI Model (\\$11.07M Impact)
 - 8.3 30-60-90 Day Executive Implementation Roadmap
 - **References**
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LIST OF TABLES

- 1. **Table 1:** Enterprise Dataset Schema & Statistical Summary
 - 2. **Table 2:** Mathematical Formulations of Engineered Career KPIs
 - 3. **Table 3:** Unsupervised Clustering Validation Benchmarks (\$K=3\$ to \$K=7\$)
 - 4. **Table 4:** Centroid Feature Profiles for 5 Career Archetypes
 - 5. **Table 5:** Promotion Gap Risk Stratification Across Departments
 - 6. **Table 6:** Manager Stability Cohorts and Promotion Latency
 - 7. **Table 7:** Prescriptive Talent Routing Rules and Intervention Matrix
 - 8. **Table 8:** Cost Avoidance Financial ROI Framework
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ABSTRACT

Traditional Human Resource (HR) analytics architectures rely primarily on binary supervised classification models designed to predict whether an individual employee will leave an organization. While predictive turnover models signal attrition risk, they operate reactively, frequently triggering alerts only after an employee has mentally disengaged, initiated external job searches, or entered terminal stages of departure.

This paper introduces **Career Intelligence as an Unsupervised Retention Strategy**. Rather than forecasting departure as an isolated binary event, this research operationalizes the underlying structural root causes of workforce disengagement: promotion velocity deficits, role tenure stagnation, managerial continuity bottlenecks, and developmental intensity gaps.

Leveraging an enterprise dataset of 1,470 employees across engineering, sales, and administration at Palo Alto Networks, we establish a mathematical framework for derived career velocity metrics and apply unsupervised machine learning (K-Means Clustering with Hierarchical agglomerative validation and Principal Component Analysis) to discover five distinct workforce career archetypes: (1) Fast-Track Performers (18.4%), (2) Stable

Long-Term Contributors (16.2%), (3) Early-Career Explorers (23.1%), (4) Promotion-Stalled Employees (27.3%), and (5) High-Risk Stagnation Profiles (15.0%).

Our findings demonstrate that **27.3% of the enterprise workforce is trapped in promotion-stalled trajectories**, and prolonged managerial continuity exceeding 6 years without role changes elevates turnover hazard by $\$3.4\times\$$. We formulate a composite **Promotion Gap Risk Score (\$0-100\$)** and a **Retention Opportunity Index (\$0-100\$)**, isolating 142 high-performing active contributors who require immediate career intervention. Finally, we deliver an interactive Streamlit decision cockpit and simulation engine projecting an annual cost avoidance of **\\$11.07 Million** through proactive talent mobility.

CHAPTER 1: INTRODUCTION AND PROBLEM FORMULATION

1.1 Background of Palo Alto Networks Enterprise Workforce

Palo Alto Networks operates in a hyper-competitive cybersecurity landscape where high-performing engineering, research, and technical sales personnel drive enterprise valuation. In knowledge-intensive cybersecurity domains, replacement costs for experienced specialized engineers average $\$1.5\times\$$ to $\$2.0\times\$$ annual salary when factoring in institutional knowledge loss, recruiting overhead, and project delay costs.

1.2 The Problem of Reactive Turnover Prediction

Existing People Analytics tools focus almost exclusively on supervised binary classification: $\hat{y} \in \{0, 1\}$ (Stay vs. Leave). This approach suffers from three fundamental limitations: 1. **Late Warning Horizon:** Predictions spike only when absenteeism, satisfaction dips, or overtime surge, leaving HR zero lead time to intervene productively. 2. **Lack of Prescriptive Mechanism:** Supervised classifiers state *that* an employee might leave, but fail to prescribe actionable career mobility solutions. 3. **Ignoring Stagnant Non-Leavers:** Employees who stay despite severe career stagnation suffer from silent productivity decay, disengagement, and quiet quitting.

1.3 Research Objectives

The primary objective of this project is to construct a **Career Intelligence Platform** that: - Formulates continuous mathematical indicators for career velocity and promotion latency. - Discovers latent career archetypes via unsupervised clustering without manual bias. - Prioritizes high-performing active talent who are experiencing promotion gaps before they disengage. - Provides counterfactual simulation to project the exact risk reduction of promotions, lateral moves, and upskilling.

CHAPTER 2: DATA ARCHITECTURE AND DOMAIN FEATURE ENGINEERING

2.1 Raw Dataset Overview

The dataset contains $N = 1,470$ employee records across 31 raw demographic, behavioral, compensation, and tenure attributes.

Workforce Baseline Statistics:	
- Total Workforce Analyzed:	1,470 Employees
- Active Employees:	1,233 (83.9%)
- Historical Attrition Cohort:	237 (16.1%)
- Mean Years at Company:	7.01 Years (Range: 0 - 40)
- Mean Years Since Last Promotion:	2.19 Years (Range: 0 - 15)
- Mean Years in Current Role:	4.23 Years (Range: 0 - 18)
- Mean Years with Current Manager:	4.12 Years (Range: 0 - 17)

2.2 Mathematical Formulations for Derived Career Indicators

To capture longitudinal career momentum, we construct five derived domain indicators with smoothing constant $\epsilon = 1.0$:

1. Promotion Gap Ratio (\$PGR\$)

Measures the proportion of company tenure spent waiting for advancement:
$$\text{PGR}_i = \frac{\text{YearsSinceLastPromotion}_i}{\text{YearsAtCompany}_i + 1.0}$$

2. Role Stagnation Index (\$RSI\$)

Measures horizontal immobility within the current job function:
$$\text{RSI}_i = \frac{\text{YearsInCurrentRole}_i}{\text{YearsAtCompany}_i + 1.0}$$

3. Training Intensity Score (\$TIS\$)

Quantifies organizational upskilling frequency normalized by tenure:
$$\text{TIS}_i = \frac{\text{TrainingTimesLastYear}_i}{\text{YearsAtCompany}_i + 1.0}$$

4. Manager Stability Indicator (\$MSI\$)

Evaluates supervisory continuity relative to role longevity:
$$\text{MSI}_i = \frac{\text{YearsWithCurrManager}_i}{\text{YearsInCurrentRole}_i + 1.0}$$

5. Career Velocity (\$CV\$)

Rates the speed of hierarchical progression across overall career span:
$$\text{CV}_i = \frac{\text{JobLevel}_i}{\text{TotalWorkingYears}_i + 1.0}$$

2.3 Composite Promotion Gap Risk Scoring (\$PGRS\$)

We formulate a continuous risk index $PGRS \in [0, 100]$:

$$\text{PGRS}_i = \min\left(100, \left(35 \cdot \min\left(1, \frac{\text{YSLP}_i}{10}\right) + 25 \cdot \min\left(1, \text{RSI}_i\right) + 25 \cdot \min\left(1, \frac{\text{YICR}_i}{8}\right) + 15 \cdot \left(1 - \frac{\text{JobSatisfaction}_i}{4}\right)\right)\right)$$

Risk Stratification: - **Low Risk (\$PGRS < 35\$):** Healthy velocity (53.4% of workforce) - **Medium Risk (\$35 \le PGRS < 60\$):** Emerging stagnation (22.3% of workforce) - **High Risk (\$PGRS \ge 60\$):** Acute career freeze (24.3% of workforce)

2.4 Retention Opportunity Index (\$ROI\$)

To operationalize proactive retention, we formulate the \$ROI \in [0, 100]\$, to prioritize **active, high-performing employees experiencing high stagnation**:

$$ROI_i = 25 \cdot (1 - Attrition_i) + 25 \cdot \left(\frac{PerformanceRating_i}{4}\right) + 15 \cdot \left(\frac{JobInvolvement_i}{4}\right) + 35 \cdot \left(\frac{PGRS_i}{100}\right)$$

CHAPTER 3: UNSUPERVISED CLUSTERING & ARCHETYPE DISCOVERY

3.1 Preprocessing Pipeline

Twelve standardized feature dimensions (\$Z\$-score standardized with $\mu=0, \sigma=1$) were fed into unsupervised algorithms: $\mathbf{x}_i = [TWY, YAC, YICR, YSLP, YWCM, JobLevel, PGR, RSI, TIS, MSI, CV, SalaryHike]$

3.2 Cluster Validation Benchmarks

Number of Clusters (\$K\$)	Silhouette Score (K-Means)	Calinski-Harabasz Index	Davies-Bouldin Index	Silhouette (Hierarchical)
\$K=3\$	\$0.204\$	\$289.4\$	\$1.72\$	\$0.188\$
\$K=4\$	\$0.211\$	\$315.2\$	\$1.64\$	\$0.195\$
\$K=5\$ (Optimal)	\$0.218\$	\$341.1\$	\$1.52\$	\$0.202\$
\$K=6\$	\$0.209\$	\$310.8\$	\$1.59\$	\$0.191\$
\$K=7\$	\$0.198\$	\$284.6\$	\$1.68\$	\$0.183\$

\$K=5\$ achieved optimal cluster cohesion, maximum Calinski-Harabasz separation (\$341.1\$), and strongest architectural alignment with real-world HR career bands.

3.3 Principal Component Analysis (PCA)

PCA reduced the 12-dimensional feature space to 3 orthogonal components capturing **\$62.83\%\$** of cumulative variance ($PC_1 = 33.4\%$, $PC_2 = 18.2\%$, $PC_3 = 11.2\%$).

3.4 Discovered Career Archetypes

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CAREER ARCHETYPE CENTROID PROFILES (K=5)

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1. Fast-Track Performers (18.4% | n=270):

- Mean Tenure: 4.8 yrs | Mean Promotion Gap: 0.8 yrs | Career Velocity: 0.38 (Highest)

- Profile: Rapid title advancement, high salary hike, low stagnation risk (14.2).

2. Stable Long-Term Contributors (16.2% | n=238):

- Mean Tenure: 19.4 yrs | Total Experience: 24.6 yrs | Job Level: 4.2

- Profile: Institutional pillars, high retention stability, deep specialized domain skills.

3. Early-Career Explorers (23.1% | n=340):

- Mean Tenure: 1.8 yrs | Total Experience: 3.1 yrs | Training Score: 1.42 (Highest)
- Profile: High learning appetite, vulnerable to rapid churn if early pathing is absent.

4. Promotion-Stalled Employees (27.3% | n=401):

- Mean Tenure: 8.9 yrs | Mean Promotion Gap: 6.4 yrs | Stagnation Score: 68.4
- Profile: High performers hitting ceiling; represents the primary turnover hazard.

5. High-Risk Stagnation Profiles (15.0% | n=221):

- Mean Role Tenure: 7.8 yrs | Role Stagnation Index: 0.88 | Satisfaction: 2.1 / 4.0
- Profile: Chronic multi-year role immobility, low engagement, high disengagement risk.

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CHAPTER 4: PROMOTION GAP & RETENTION OPPORTUNITY MODELING

Cross-departmental analysis reveals significant variation in stagnation risk: - **Research & Development:** \$23.1\%\$ high promotion gap risk; primary bottleneck occurs between Job Level 2 (Research Scientist) and Level 3 (Senior Scientist). - **Sales:** \$28.4\%\$ high promotion gap risk; average role tenure before promotion is \$4.8\$ years vs. \$2.8\$ years market benchmark. - **Human Resources:** \$16.7\%\$ high promotion gap risk; highest stability index.

The algorithm identifies **142 active high-performing employees** in the Immediate Action retention queue (\$ROI \ge 75.0\$) who are performing at above-average ratings but have not received a promotion in \$>4.0\$ years.

CHAPTER 5: MANAGERIAL CONTINUITY & LEADERSHIP IMPACT

5.1 Managerial Tenure vs. Promotion Latency

Empirical analysis demonstrates a non-linear relationship between manager tenure and career progression:

Manager Continuity Cohorts:

-
- | | |
|------------------------------|--|
| - 0 to 2 Years with Manager: | Mean Promo Gap: 1.2 yrs Attrition: 19.8% (Transition Shock) |
| - 2 to 5 Years with Manager: | Mean Promo Gap: 2.1 yrs Attrition: 11.4% (Optimal Growth Zone) |
| - 6+ Years with Manager: | Mean Promo Gap: 4.9 yrs Attrition: 24.6% (Stagnant Dyad) |
-

Employees remaining under the same direct supervisor for \$>6\$ years without role changes experience a **\$2.3\times\$ increase in promotion latency** and a **\$3.4\times\$ increase in turnover hazard**.

CHAPTER 6: INDIVIDUAL CAREER SIMULATOR & WHAT-IF LAB

The platform integrates a counterfactual simulation engine that allows HR business partners and people managers to project intervention outcomes:

$$\mathbf{x}_{\text{sim}} = f(\text{Promotion}, \text{Lateral Move}, \text{Training}, \text{Manager Rotation})$$

When an intervention is simulated: 1. Recomputes derived feature vector $\mathbf{x}_{\text{sim}}$. 2. Projects $\mathbf{x}_{\text{sim}}$ into the PCA latent coordinate space: $\mathbf{z}_{\text{sim}} = \mathbf{x}_{\text{sim}} \mathbf{W}_{\text{PCA}}$. 3. Computes Euclidean distance to centroid clusters and predicts new archetype membership. 4. Calculates instantaneous risk mitigation $\Delta \text{PGRS} = \text{PGRS}_{\text{sim}} - \text{PGRS}_{\text{baseline}}$.

Empirical Simulation Example (PANW-1001): - Baseline State: Promotion Gap Risk = \$48.8\$, Archetype = *High-Risk Stagnation Profile*. - Action Simulated: *Award Promotion (Reset YSLP to 0, Increment Job Level)*. - Simulated State: Promotion Gap Risk drops to **\$13.8\$ (\$-35.0\$ risk reduction)**, and archetype transitions to **Fast-Track Performer**.

CHAPTER 7: STREAMLIT ENTERPRISE DECISION COCKPIT

The web application is engineered with an enterprise Slate and Cyan design system (#0B0F17 dark canvas, #111827 cards, #0EA5E9 accents) with 8 integrated analytical modules: 1. **Executive Overview:** High-level workforce health indicators and archetype distribution. 2. **Career Path Clustering:** Interactive 2D/3D PCA visualizations and radar archetype profiles. 3. **Promotion Gap Monitor:** Cross-departmental stagnation heatmaps and tenure scatter plots. 4. **Retention Opportunity Panel:** Prescriptive intervention roster with CSV export. 5. **Managerial & Leadership Impact:** Supervisory tenure vs promotion latency analysis. 6. **Career Simulator & What-If Lab:** Counterfactual scenario modeling. 7. **Workforce Data Explorer:** Granular employee lookup and filtering. 8. **Research Documentation & Executive Policy Briefing:** Full research paper and executive briefing with PDF download hooks.

CHAPTER 8: CONCLUSION, ROI MODEL & POLICY RECOMMENDATIONS

8.1 Summary of Quantitative Impacts

- **\$27.3%\$ of workforce** identified in promotion-stalled trajectories before initiating exit behavior.
- **142 high-performing contributors** isolated for immediate proactive retention routing.
- **\$3.4\times\$ turnover hazard** linked to stagnant supervisory dyads (>6 years).

8.2 Cost Avoidance Financial ROI Model

$$\text{Annual Cost Avoidance} = N_{\text{queue}} \times P_{\text{baseline attrition}} \times E_{\text{intervention efficacy}} \times C_{\text{replacement}}$$

$$\text{ROI} = 142 \times 0.65 \times 0.75 \times \$160,000 = \mathbf{\$11,076,000 \text{ Annually}}$$

8.3 30-60-90 Day Executive Action Roadmap

30-DAY IMMEDIATE ACTIONS (TRIAGE):

- Deploy Retention Opportunity Panel to HR Business Partners.

- Initiate immediate Career Progression Reviews for all 142 employees in the High-Priority Queue.
- Flag all 89 stagnant manager dyads (>6 years) for lateral rotation discussions.

60-DAY STRUCTURAL INTEGRATION:

- Embed the 24-month promotion latency trigger into Workday/HRIS compensation cycles.
- Launch cross-functional rotation tracks in Research & Development for Level 2 Scientists.
- Institute mandatory manager mentorship reassignment protocols at the 4-year mark.

90-DAY POLICY & GOVERNANCE:

- Tie managerial quarterly performance incentives to team career mobility velocity.
 - Formalize technical fellowship ladders for Stable Long-Term Contributors.
 - Review annual cost avoidance against baseline turnover metrics.
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REFERENCES

1. Allen, D. G., Bryant, P. C., & Vardaman, J. M. (2010). *Retaining Talent: Replacing Misconceptions with Evidence-Based Strategies*. *Academy of Management Perspectives*, 24(2), 48–64.
2. Campion, M. A., Cheraskin, L., & Stevens, M. J. (1994). *Career-Related Antecedents and Outcomes of Job Rotation*. *Academy of Management Journal*, 37(6), 1518–1542.
3. Griffeth, R. W., Hom, P. W., & Gaertner, S. (2000). *A Meta-Analysis of Antecedents and Correlates of Employee Turnover*. *Journal of Management*, 26(3), 463–488.
4. MacQueen, J. (1967). *Some Methods for Classification and Analysis of Multivariate Observations*. *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*, 1, 281–297.
5. Ward, J. H. (1963). *Hierarchical Grouping to Optimize an Objective Function*. *Journal of the American Statistical Association*, 58(301), 236–244.

Report submitted to Unified Mentor by Sarupya Guha (Intern ID: UMID150826105831) — Human Resources Analytics & Career Intelligence.